

Internet of Things (IoT)

Chapter 2

IoT Ecosystem

Different components of IoT

Course Code: CSE0612405

"Strength is Life, Weakness is Death." – Swami Vivekananda

IoT Ecosystem

It encompasses **all the components** needed to collect, process, and analyze data from IoT devices, enabling smart applications and services.

✓ IoT Core

- Sensors & Actuators, microcontrollers, **internet connectivity**, service platform including **security**

✓ IoT Gateway

- It ensure bidirectional communication between IoT networks and other networks

✓ User Interface / Visualization

- Design sleek, visually appealing, interactive, and ease-of-use graphical user interface (GUI)

✓ IoT Architectures

- **Graphical structure** of the designed IoT-based solutions and products
 - 3-layer architecture, IoT-WF, oneM2M, etc.

✓ Cloud

- Accepts, accumulates, maintains, stores, and process data

✓ Analytics

- It indulges in conversion and **analysis of data** which results in recommendations and future decision making

IoT Ecosystem

It encompasses **all the components** needed to collect, process, and analyze data from IoT devices, enabling smart applications and services.

✓ IoT Frameworks

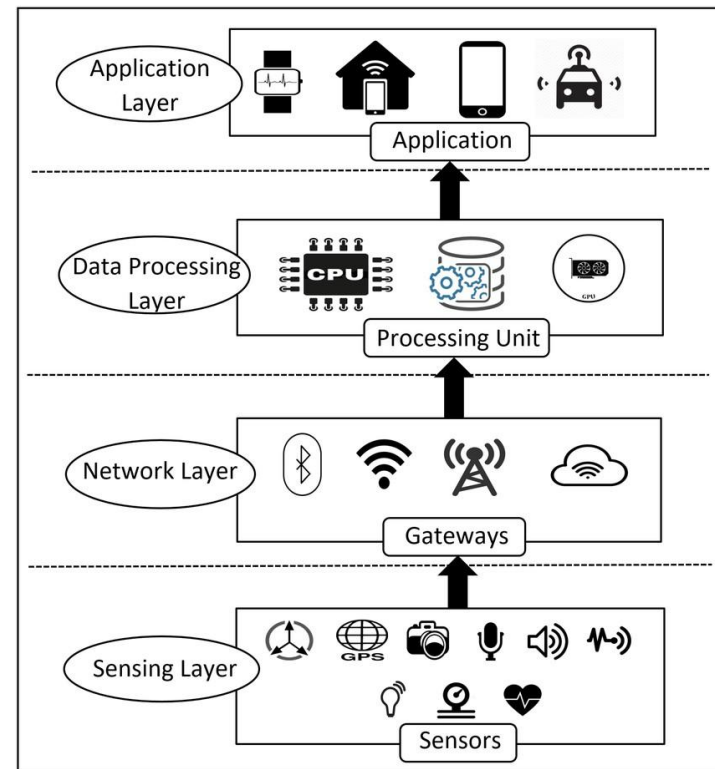
- Tools needed to design and implement IoT-based solutions and products
 - Microsoft Azure IoT,
 - Cisco Ultra IoT,
 - Amazon AWS IoT,
 - IBM Watson IoT,
 - etc.

IoT Network Architecture

- Network and its application should never be built without careful planning
- Architecture is how you design (i.e. [graphical structure](#)) your solution.

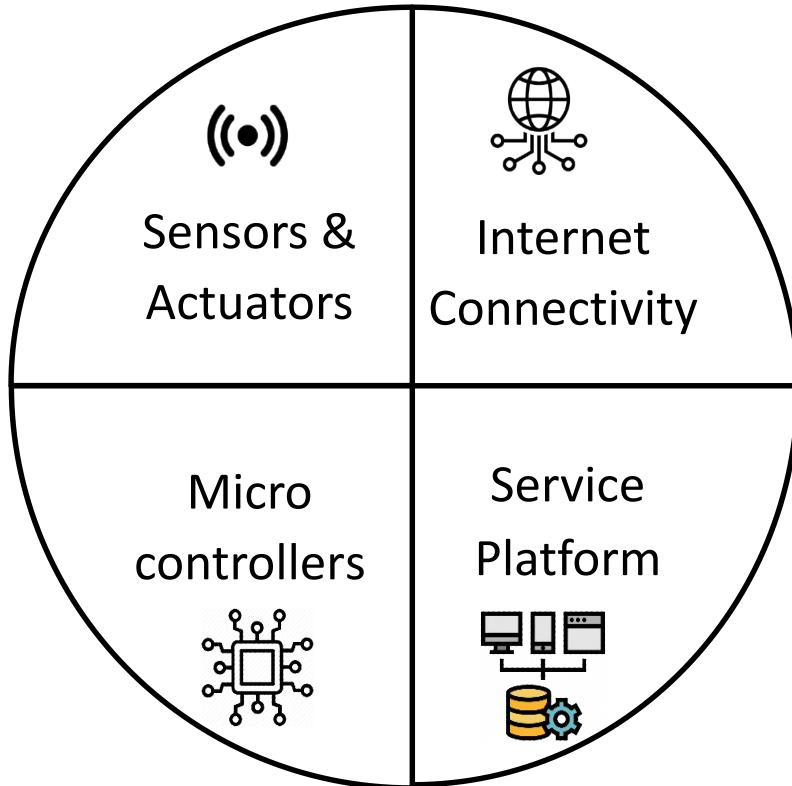
- IoT architecture refers to the framework that defines [how](#) different components of an IoT system [interact](#) to enable data collection, [transmission](#), [processing](#), and ultimately the [delivery of services](#).

Very common 4-Layer Architecture for IoT



Source: Sikder *et al.*, "A Survey on Sensor-based Threats to Internet-of-Things (IoT) Devices and Applications" *IEEE Communications Surveys & Tutorials*, 23(2), pp. 1125 - 1159, 2021.

Core Components of IoT



- **Sensors** - to gather data and events
- **Actuators** – responsible for moving and controlling a mechanism or system
- **Microcontrollers** - automatically controls sensors and actuators; makes them smart
- **Internet connectivity** – responsible for **sharing information** and control command
- **Service Platform** – ability to **deploy and manage the IoT devices and applications** including data management, data analytics and **security**

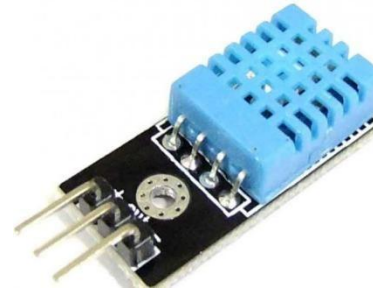
“Things” in IoT – Sensors



MQ135 - Air Quality Gas Sensor



Sound Detection Sensor



DHT11 - Temperature and Humidity Sensor



PIR Motion Detector Sensor



Pulse Sensor



LDR Light Sensor



Ultrasonic Distance Sensor



IR Sensor

“Things” in IoT – Actuators



4 Channel 5V Relay



Servo Motor



DC Motor



Solenoid valve



Linear Actuators



LED



LCD Display

Access Technologies in IoT

Communication Criteria

- Range
- Frequency Bands
- Power Consumption
- Topology
- Constrained Devices
- Constrained-Node Networks

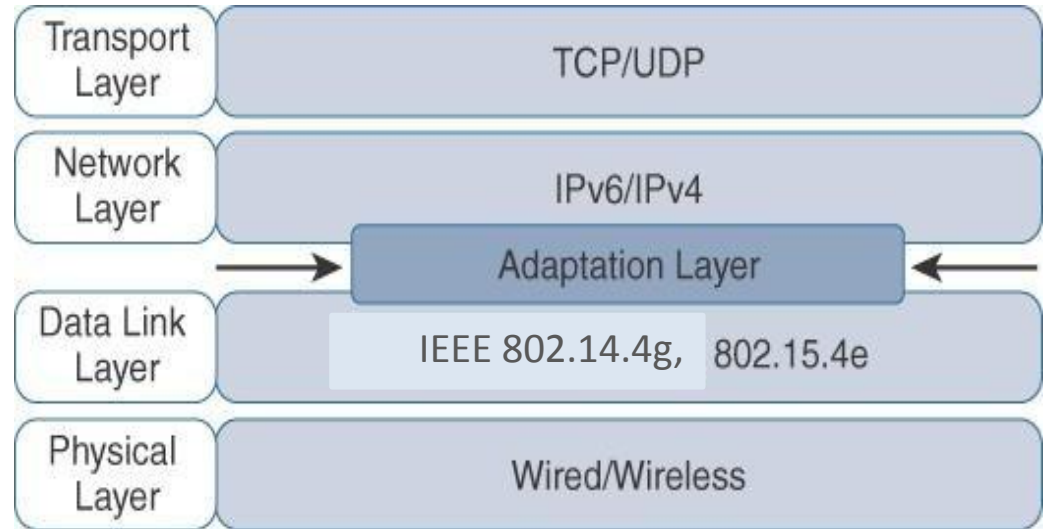
IoT Access Technologies



Use of Internet Infrastructure

Key Advantages of IP

- ❑ Open and standard-based
- ❑ Versatile
- ❑ Ubiquitous
- ❑ Scalable
- ❑ Manageable
- ❑ Highly secure
- ❑ Stable and resilient



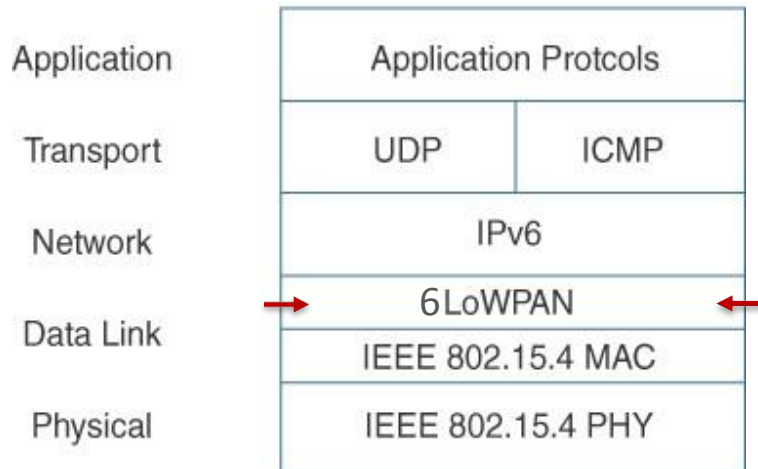
- IPv6 packets require a minimum MTU/PDU size of **1280 bytes**.
- The maximum size of a MAC layer frame in IEEE 802.15.4 is **127 bytes**.
 - It gives just **102 bytes** for an **IPv6 packet** !!

Need of packet/frame size optimization due to

- ❑ Constrained Nodes
- ❑ Constrained Networks

Modification in Stack

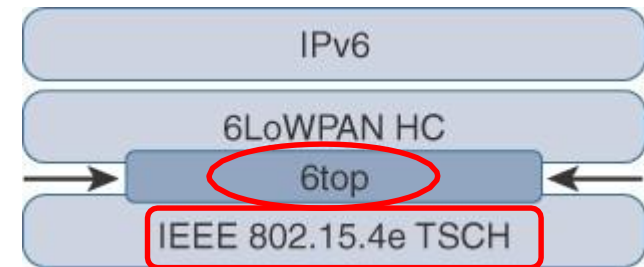
IoT Protocol Stack with
6LoWPAN Adaptation Layer



IEEE 802.15.4e -2011
Amendment

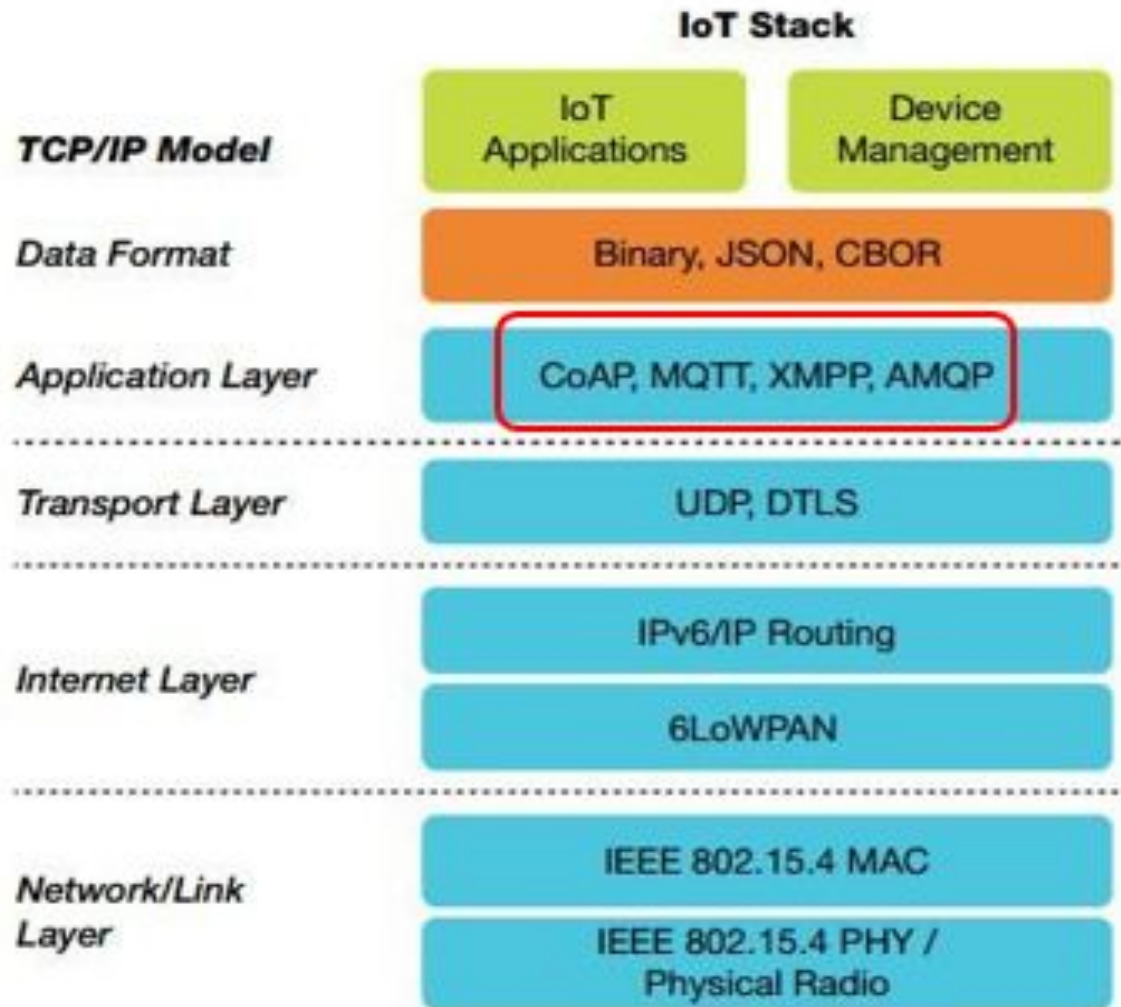


In 6TiSCH IoT Network



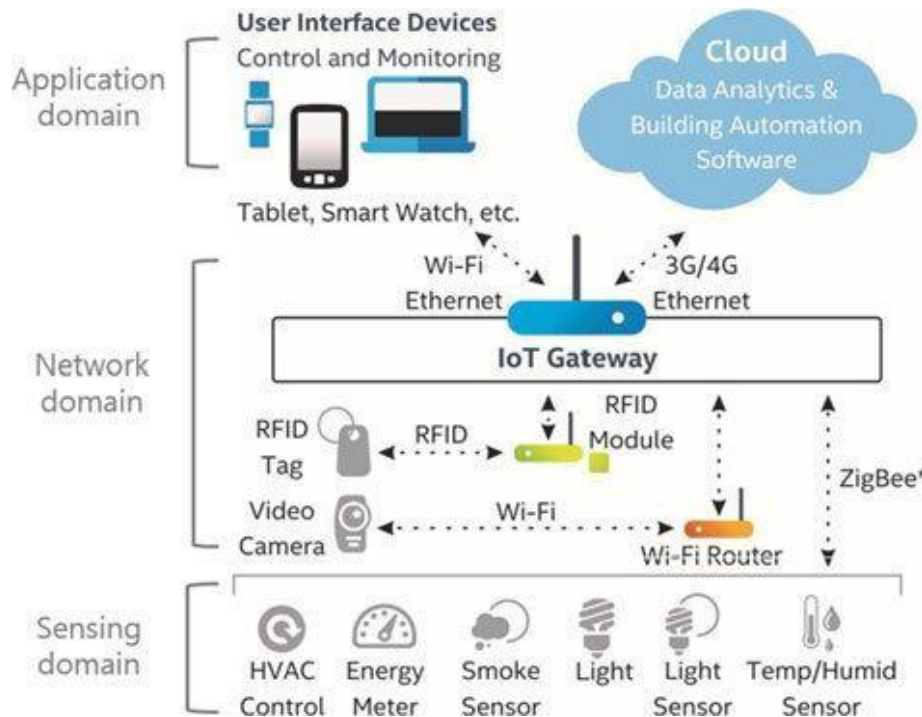
- **WPAN**: Wireless Personal Area Networks
- **IEEE 802.15.4**: Low-Rate WPAN
- **6LoWPAN**: IPv6 over Low-Power WPAN
- **TSCH**: Time Synchronized Channel Hopping
- **6TiSCH**: IPv6 over the TSCH mode of IEEE 802.15.4e
- **6top**: 6TiSCH Operation Sublayer

Application Layer



IoT Gateway

- It is a **physical device or software program** that serves as the connection point between the two different types of networks



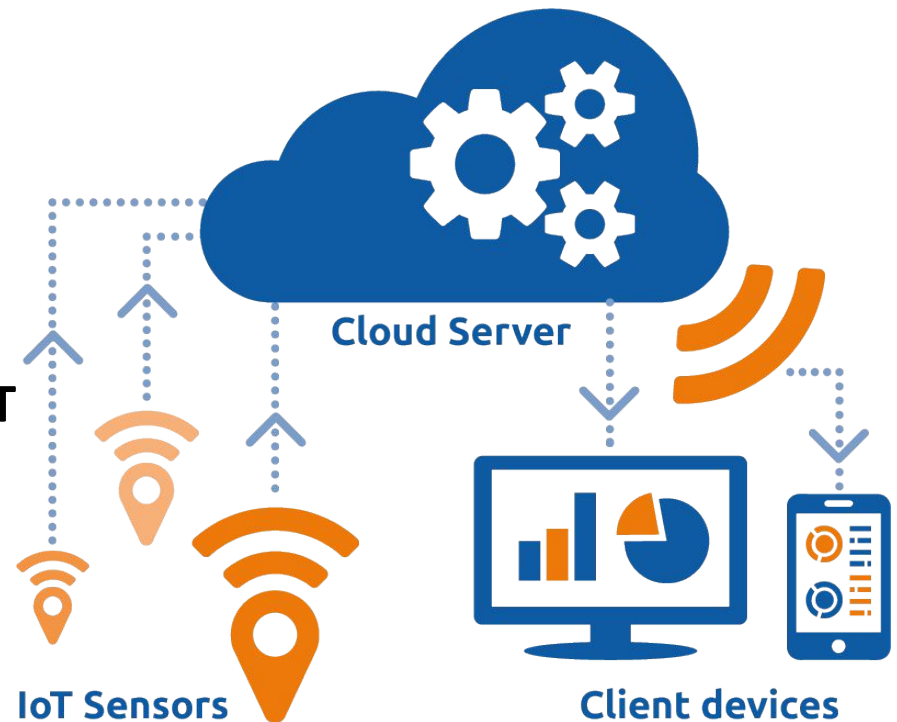
Gateway

- Provides bidirectional **communication**
 - Between IoT protocols and other networks
 - e.g. Zigbee <--> Ethernet
- Sometimes programmed to execute some **processing** operations
 - Edge computing
- It is necessary to maintain **security** to a certain extent
 - Can shield the entire IoT systems from any cyberattack

Source: B. Kang, D. Kim, H. Choo, "Internet of Everything: A Large-Scale Autonomic IoT Gateway", IEEE Transactions on Multi-scale Computing Systems, vol. 3, no. 3, 2017, pp. 206-214.

Use of Cloud

- IoT generates **vast amount of Big Data**; this in turn puts a huge strain on Internet Infrastructure.
- **Cloud can facilitate** to
 - Provide different **services**
 - **Store** huge amount of data
 - **Process** the data efficiently
- **Benefits of Cloud Platform in IoT**
 - Network Scalability
 - Data Mobility
 - Time to market
 - Security
 - Cost-effectiveness

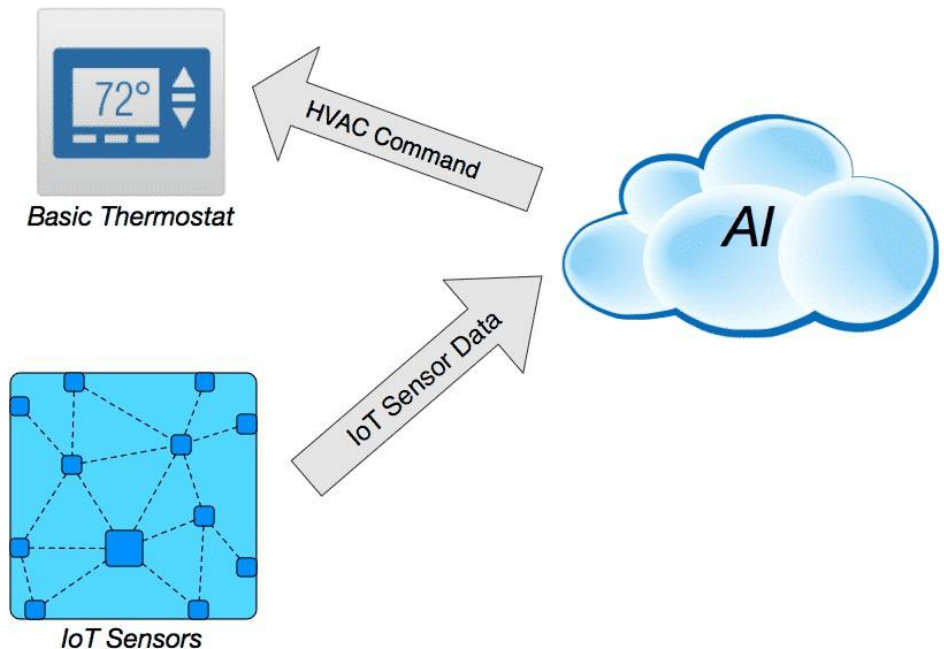


AI for IoT

- AI focuses on **putting human intelligence in machine**
- It gives the ability to a machine/program to **think and learn by itself**

Use of AI in IoT:

- **Smart Home**
 - Automated HVAC control
- **Industrial IoT**
 - Predictive maintenance
 - Optimized supply chain
- **Farming**
 - Smart farming
 - Interruption warning
- **Self-driving Car**
 - Mimic human driving on road
- **Health**
 - Auto-diagnosing any disease
 - Assistive healthcare

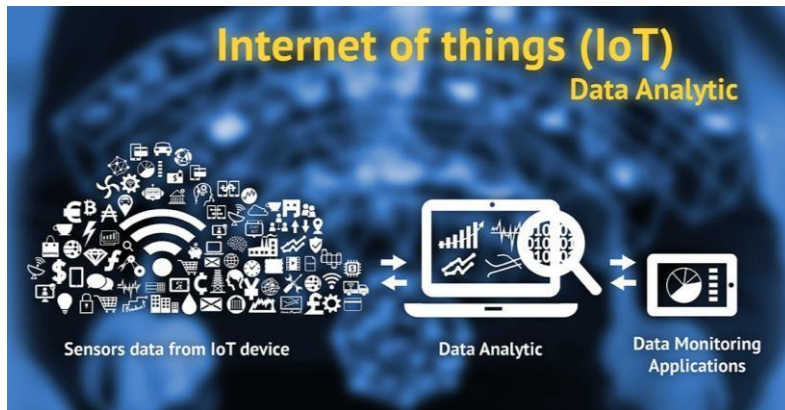


Data Analytics in IoT

- ❖ The **business value of IoT** is not just in the ability to **connect devices**, but it comes from **understanding the data** these devices create.

“Data Analytics + IoT => Smart Business Solutions”

- **IoT analytics** is the **application of data analysis tools and procedures** to realize value from the huge volumes of data generated by connected IoT devices



Challenges:

- ✓ Huge **Volume**
- ✓ Real-time data **flow**
- ✓ **Variety** of data types
 - e.g. XML, video, SMS
- ✓ Unstructured data
- ✓ **Variable** data model and meaning / **value**

Securing IoT

- Both the IoT **manufacturers** and their **customers** didn't care about the security !

Unauthorized access to IoT devices



Source: <https://www.theguardian.com/technology/2016/oct/26/ddos-attack-dyn-mirai-botnet>

Major cyber attack disrupts internet service across Europe and US;
October 26, 2016

Unauthorized access to IoT network



Source: <http://metropolitan.fi/entry/ddos-attack-halts-heating-in-finland-amidst-winter>

DDoS attack halts heating in Finland amidst winter;
November 7, 2016

User Interface

- Information made available to the end-users
- Users can actively **check and act in** for their IOT system



Important Characteristics:

- ✓ Sleek design
- ✓ Visually appealing
- ✓ Interactive UI
- ✓ Ease-of-use
- ✓ Handy

Source: <https://www.daikin.com/about/design/2017/05/entry-15>

IoT Framework

- Framework provides a **development environment**.
 - It provides appropriate infrastructure **to design and implement the architecture**
- IoT framework comprises of **large number of components**
 - sensors, sensor systems, gateways, mobile app, embedded controller, data management platform, analytical platform, and so on.
 - support **interoperability** among all devices, provides **secure connectivity**, **reliability** in data transfer, **interface** to 3rd party application to built on it, and so on.

Few IoT Framework	Few IoT Framework
RTI (Real-Time Innovations) Connex DD	Cisco Ultra IoT
Salesforce IoT cloud	Microsoft Azure IoT
Eclipse IoT	PTC ThingWorx
GE (General Electronic) Predix	Amazon AWS IoT
IBM Watson IoT	Kaa

Lessons Learned

- What is IoT Ecosystem

- Different components of IoT
 - IoT Architecture
 - IoT Core
 - ✓ Sensors & Actuators
 - ✓ IoT Access Technologies
 - ✓ Use of Internet
 - IoT Gateway
 - Use of Cloud in IoT
 - AI for IoT
 - Data Analytics in IoT
 - Security in IoT
 - User Interface for IoT
 - IoT Framework

Thanks!

